



Developments in IMU and Acoustic Swallow Detection

Objectives:

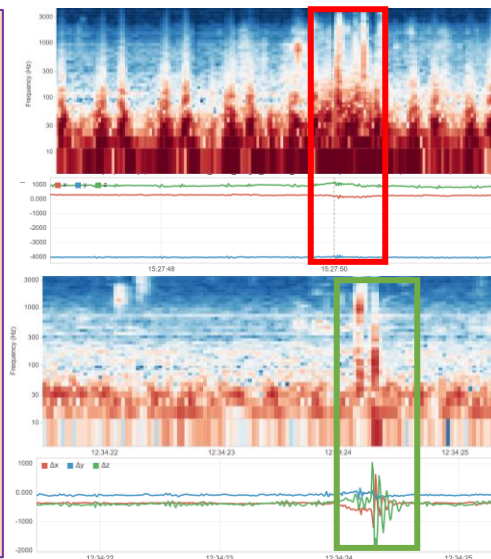
Acoustic Swallow Detection

- Improve ambulatory swallow detection using IMU + Mic
- Develop a deployable clinical testing protocol
- Compare Single vs. Dual IMU for detection during rest, movement and mealtime

R01 Swallow Annotation

- ~500 verified markers placed across 45 datasets
- Time alignment for clock skew

Resting Swallows (single, dual)

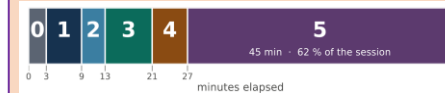


Technical Approach:

Acoustic Swallow Detection

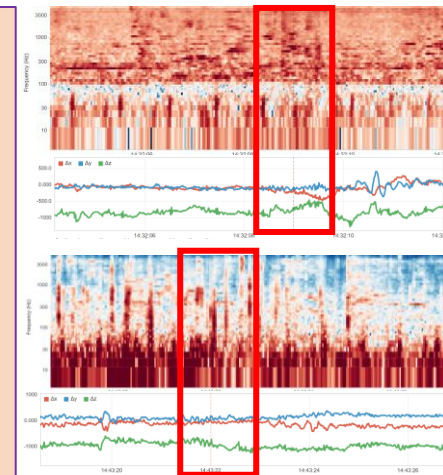
- 72-minute protocol:

§0: Biometrics §1: Vocalizations
§2: IMU calibration §3: Seated Bolus
§4: Moving Bolus §5: Ambulation



- Specificity & Sensitivity over Accuracy for swallow detection
- Sensitivity: True swallow detection
- Specificity: False swallow rejection

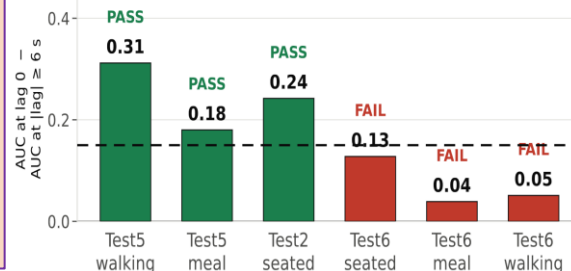
Noisy, Eating Swallows



Technical Accomplishments:

Acoustic Swallow Detection

- AUC method evaluating clinician markers
- 20% Dual IMU, 80% Mic threshold weighting (log)



Session	Sensitivity	Specificity	FP/Hour
Resting	38%	86%	11.1
Walking	17%	24%	59.6
Eating	43%	41%	173.2

Design and Data, Future, Applications:

Acoustic Swallow Detection

Future of Sensor (Mic + IMU)

- Clear Resting Swallows
- Confounded by chews & motion

Future of Protocol

- Captures all event types
- Create new swallow Baseline
- Testing dot tracking + depth sensors

Fig. a, q taken from Kang 2022

